Lab 10 — Monitoring & Observability — Prometheus, Grafana, Loki

ID: K8S-LAB-10 · **Category:** Enterprise · **Level:** Enterprise · **Duration:** 75 min

Focus: Prometheus, Grafana, Loki, metrics-server, alerting

Objective

Install the kube-prometheus-stack, visualize cluster metrics in Grafana, ship logs to Loki, and write a basic PromQL alert.

Concepts Covered

• Pull-based metrics vs push

• Service discovery via ServiceMonitor

• Log labels vs full-text indexing

Prerequisites

• A cluster from Lab 1

• Helm 3 installed

Architecture

Prometheus scrapes; Loki indexes labels not log lines (cheaper); Grafana stitches both into a single dashboard.

```
Pods --(/metrics)--> Prometheus  ---->  Grafana
            (kubelet/cadvisor)
Pods --(stdout)--->  Promtail  ---->  Loki  ---->  Grafana
```

Lab Steps

01. Install kube-prometheus-stack

One chart ships Prometheus, Grafana, Alertmanager, node-exporter and a pile of dashboards.

```
helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
helm repo update
helm install monitoring prometheus-community/kube-prometheus-stack -n monitoring --create-namespace
```

Expected: All pods Running 1/1 in `monitoring` within ~2 min.

Internals: ServiceMonitor CRDs are how Prometheus discovers what to scrape inside the cluster.

02. Open Grafana

Default admin password is `prom-operator` for the chart.

```
kubectl -n monitoring port-forward svc/monitoring-grafana 3000:80
# in browser: http://localhost:3000 (admin / prom-operator)
```

Expected: Grafana UI loads with prebuilt 'Kubernetes / Compute Resources' dashboards.

Internals: Port-forward tunnels via the API server; no LoadBalancer needed for the lab.

03. Install Loki and Promtail

Promtail (DaemonSet) ships /var/log/containers/* to Loki; Grafana already discovers Loki via the same release.

```
helm install loki prometheus-community/loki-stack -n monitoring
```

Expected: loki-* pods Running.

Internals: Promtail tails container log files and adds Pod labels as Loki labels.

04. Write a PromQL alert

Fires when any pod restarts more than 6×/hour for 5 minutes.

```
cat <<EOF | kubectl -n monitoring apply -f -
apiVersion: monitoring.coreos.com/v1
kind: PrometheusRule
metadata: {name: pod-restarts, labels: {release: monitoring}}
spec:
  groups:
  - name: pods
    rules:
    - alert: PodRestartingTooMuch
      expr: rate(kube_pod_container_status_restarts_total[5m]) > 0.1
      for: 5m
      labels: {severity: warning}
      annotations: {summary: "Pod {{ \$labels.pod }} restarting"}
EOF
```

Expected: prometheusrule.monitoring.coreos.com/pod-restarts created.

Internals: The Prometheus Operator reloads Prometheus when PrometheusRule CRs change.

Common Errors

No data in Grafana — Cause: ServiceMonitors filtered by `release` label mismatch · Fix: Add `labels: {release: monitoring}` to your ServiceMonitor

Loki disk fills fast — Cause: Default retention too long for log volume · Fix: Tune `limits_config.retention_period`

Troubleshooting

• `kubectl -n monitoring get servicemonitor`

• Prometheus → Status → Targets to see who's being scraped

Enterprise Relevance

You cannot run Kubernetes blind. The stack above is the lingua franca; nearly every SRE job description assumes it.

Validation

• You can see CPU/memory per Pod in Grafana

• You can query logs by Pod label in Loki

• You wrote one alert

Related Incidents & Guide Sections

Incidents: K8S-INC-3001, K8S-INC-3004

• Observability stack

Questions

Beginner

Q. Why pull instead of push for metrics?

A. Easier service discovery and built-in liveness signal.

Intermediate

Q. What's a ServiceMonitor?

A. A CRD the Prometheus Operator turns into a scrape config.

Advanced

Q. Why does Loki index labels not log content?

A. Drastically lower storage and cost; you grep within the time-bounded label stream.

Scenario

Q. Prometheus disk fills every weekend. What do you tune?

A. Retention, scrape interval, drop noisy series with relabel rules.

Viva

Q. Difference between recording and alerting rules?

A. Recording precomputes a series; alerting fires when an expr stays true for `for`.